

Ex. No: 13 File Organization Techniques

PROGRAM A: SEQUENTIAL FILE ORGANIZATION

C PROGRAM

```
#include <stdio.h>

struct student
{
    int regno;
    char name[20];
};

int main()
{
    FILE *fp;
    struct student s;
    fp = fopen("student.dat", "w");
    printf("Enter Register Number: ");
    scanf("%d", &s.regno);
    printf("Enter Name: ");
    scanf("%s", s.name);
    fprintf(fp, "%d %s\n", s.regno, s.name);
    fclose(fp);
    fp = fopen("student.dat", "r");
    fscanf(fp, "%d %s", &s.regno, s.name);
    printf("\nRecord Details\n");
    printf("Register Number : %d\n", s.regno);
    printf("Name : %s\n", s.name);
    fclose(fp);
    return 0;
}
```

Output:

```
(kali㉿kali)-[~]
└─$ nano sequential.c
└─$ gcc sequential.c -o sequential
└─$ ./sequential
Enter Register Number: 101
Enter Name: Arun

Record Details
Register Number : 101
Name : Arun
```

Register Number :
Name : Arun

Step 6: Check the crea
The program creates (st
Ask anything

SHELL SCRIPT

```
#!/bin/bash

echo "Enter Register Number:"

read regno

echo "Enter Name:"

read name

echo "$regno $name" > student.txt

echo "Contents of File"

cat student.txt
```

Output:

```
(kali㉿kali)-[~]
└─$ chmod +x sequential.sh
└─$ ./sequential.sh
Enter Register Number:
102
Enter Name:
Ajay
Contents of File
102 Ajay
```

PROGRAM B: DIRECT (RANDOM) FILE ORGANIZATION

C PROGRAM

```
#include <stdio.h>

struct student
```

```
{  
int regno;  
char name[20];  
};  
int main()  
{  
FILE *fp;  
struct student s;  
fp = fopen("random.dat", "wb+");  
printf("Enter Register Number: ");  
scanf("%d", &s.regno);  
printf("Enter Name: ");  
scanf("%s", s.name);  
fwrite(&s, sizeof(s), 1, fp);  
rewind(fp);  
fread(&s, sizeof(s), 1, fp);  
printf("\nRecord Found\n");  
printf("Reg No : %d\n", s.regno);  
printf("Name : %s\n", s.name);  
fclose(fp);  
return 0;  
}
```

Output:

```
(kali㉿kali)-[~]
└─$ nano random.c
Enter Register Number: 103
Enter Name: Vijay
Record Found
Reg No : 103
Name : Vijay
```

SHELL SCRIPT

```
#!/bin/bash

echo "Enter Record:"

read rec

echo "$rec" > random.txt

echo "Random Access Record"

sed -n '1p' random.txt

Output:
```

```
(kali㉿kali)-[~]
└─$ nano random.sh
Enter Record:
103 Vijay
Random Access Record
103 Vijay
```

PROGRAM C: INDEXED FILE ORGANIZATION

C PROGRAM

```
#include <stdio.h>

struct student
{
    int regno;
```

```
char name[20];

};

int main()
{
    struct student s[3];
    int key, i;
    printf("Enter 3 Student Records\n");
    for(i=0;i<3;i++)
    {
        scanf("%d %s",&s[i].regno,s[i].name);
    }
    printf("Enter Register Number to Search: ");
    scanf("%d",&key);
    for(i=0;i<3;i++)
    {
        if(s[i].regno==key)
        {
            printf("\nRecord Found\n");
            printf("Reg No : %d\n",s[i].regno);
            printf("Name : %s\n",s[i].name);
            return 0;
        }
    }
    printf("Record Not Found\n");
    return 0;
}
```

Output:

```
103 vijay
(kali㉿kali)-[~]
$ nano indexed.c
Then enter the
Enter number of processes:
Enter Register
Expected output
Record Found
Reg No : 102
Name : Kuma
Ask anything
```

SHELL SCRIPT

```
#!/bin/bash
```

```
echo "Enter Student Records"
```

```
echo "101 Arun" > index.txt
```

```
echo "102 Kumar" >> index.txt
```

```
echo "103 Ravi" >> index.txt
```

```
echo "Enter Register Number to Search:"
```

```
read key
```

```
grep "^$key" index.txt
```

Output:

```
(kali㉿kali)-[~]  
$ nano indexed.sh  
  
(kali㉿kali)-[~]  
$ chmod +x indexed.sh  
  
(kali㉿kali)-[~]  
$ ./indexed.sh  
Enter Student Records  
Enter Register Number to Search:  
102  
102 Kumar
```